

Analysis First-Year Exam, August 2015, Part 1

Do exactly 4 problems. Work must be presented in a neat and logical fashion in order to receive credit. Do not leave any gaps. When a theorem is used in a proof it must be precisely stated.

1. Let E be a dense subset of a metric space X and let U be an open subset of X . Prove that $U \subset \overline{E \cap U}$. (Here $\overline{E \cap U}$ denotes the closure of $E \cap U$.)
2. Let X be a metric space and $K \subset X$ a compact subset. Prove that K is closed.
3. Let X be a compact metric space and $\mathcal{F}$ a collection of closed subsets of X with the following property: for all finite subcollections $\mathcal{G} \subset \mathcal{F}$, the intersection $\bigcap_{F \in \mathcal{G}} F$ is nonempty. Prove that $\bigcap_{F \in \mathcal{F}} F$ is nonempty.
4. Let X, Y be metric spaces and $f : X \rightarrow Y$ a continuous function. Prove that if X is connected, then its image $f(X)$ is connected.
5. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers and σ a permutation of $\mathbb{N}$. For each of the following provide either a proof or a counterexample.
 - a) If the sequence $(a_n)_{n \in \mathbb{N}}$ converges then so does the rearrangement $(a_{\sigma(n)})_{n \in \mathbb{N}}$.
 - b) If the series $\sum_{n \in \mathbb{N}} a_n$ converges then so does the rearrangement $\sum_{n \in \mathbb{N}} a_{\sigma(n)}$.
6. Let $f(x) = x^2 \sin \frac{1}{x}$ for $x \in \mathbb{R} \setminus \{0\}$ and $f(0) = 0$. Prove that f is differentiable at every $x \in \mathbb{R}$. Is f' continuous at $x = 0$? (Prove your claim.)